

Technology Stack

Date	27 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

Architecture Component/ Layer	Technology Chosen	Purpose	Justification
Frontend/ Client side	React.js +Vite, HTML5, CSS3,	Develop an interactive web interface for audio upload, evaluation results, and dashboard.	Streamlit enables rapid development of AI applications with an easy-to-use and responsive interface.
Backend	Python 3.12	Implement application logic, audio processing, semantic analysis, report generation, and workflow management.	Python provides extensive support for AI, NLP, and audio processing libraries.
Speech-to-Text Engine	OpenAI Whisper	Convert users' spoken explanations into accurate text transcripts.	Whisper provides highly accurate multilingual speech recognition and robust

			transcription performance.
Semantic Analysis	Sentence-BERT (SBERT)	Compare the transcript with the reference concept and calculate semantic similarity.	SBERT generates meaningful sentence embeddings for accurate concept understanding evaluation.
Audio Analysis	Librosa	Extract speech features such as pause ratio, RMS energy, speech rate, and filler word analysis.	Librosa is a powerful Python library designed for audio signal processing and feature extraction.
Third-Party APIs	Whisper, Sentence-Transformers, Librosa, NumPy, Pandas, Matplotlib	Perform transcription, semantic analysis, audio processing, data manipulation, and visualization.	These libraries provide the core AI, NLP, signal processing, and visualization capabilities required for VBCUA.